23/9/25 Dr. Nitish Mohan 9410194181.

Roll No. | | | | | | | |

Paper Code: MGI-332

M.Tech. (Geo-Informatics)

Mid Semester Examination 2025

III *Semester*

Application of Geo- informatics in Water Resources Management

Time: 1:30 Hours MM: 50

Note:

(i) All questions are compulsory.

(ii) Answer any one sub questions among a & b in each main question.

(iii) Total marks for each main question are ten.

Q1 (1x10 = 10 Marks)

a Explain the terms Population and sample, Central tendency and Variability in elements of Statistics

b "GIS functions for spatial data capture include the numerous technologies for data capture as well as the many ways for conversion of source data into GIS-compatible formats" explain all the functions. CO1

Q2 (1x10 = 10 Marks)

a Describe Node, Arc and Polygon and differentiate between the Vector and Raster Data Structures

b Explain the function of hydrology in water resources development? Explain the basic hydrological requirements for a river basin development CO1

Q3 (1x10 = 10 Marks)

a "Precipitation products are estimated ground-accumulated rainfall. Precipitation products available include" enlist and describe the available form of data

b Describe Estimation Of Water-Supply Demands and enlist Water-Supply and Irrigation Systems Data. CO2

Q4

a "Satellite imagery can provide useful information on rainfall distribution over large areas and inaccessible regions," describe the estimation form

b Describe GIS-Based Water Network Optimization and draw the flowchart of WADSOP optimization procedure. CO2

Q5 (1x10 = 10 Marks)

a "GIS program has a number of terrain-processing routines that can be integrated into computer programs" enlist all the terrain-processing routines CO2

b Explain in detail all the Soil-Moisture and Infiltration Models